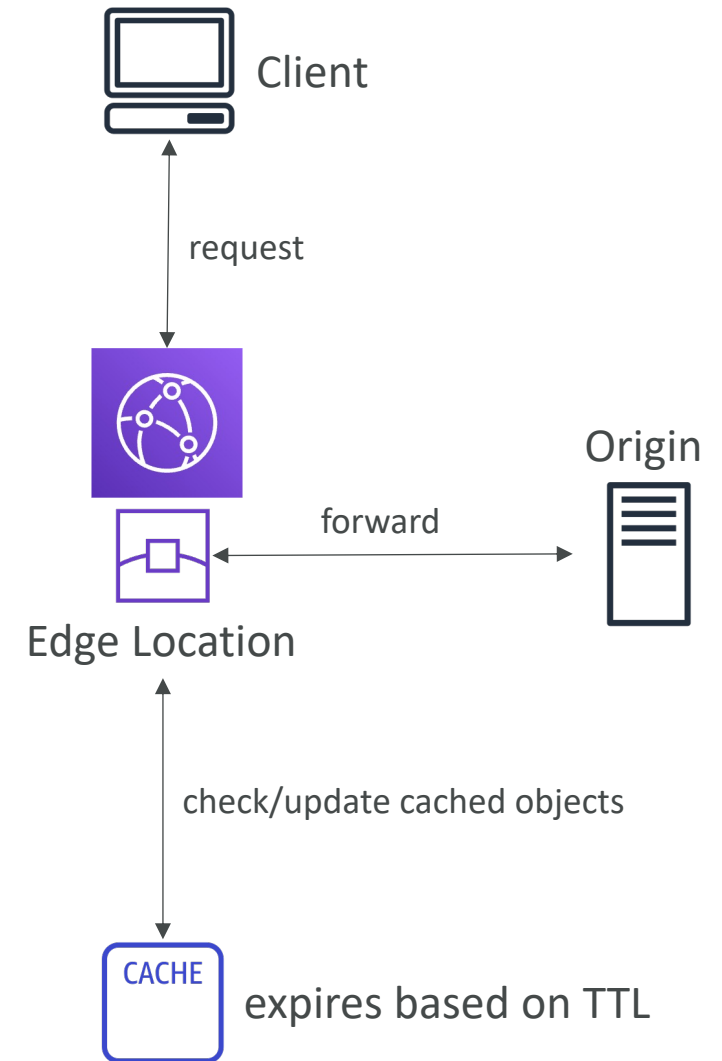


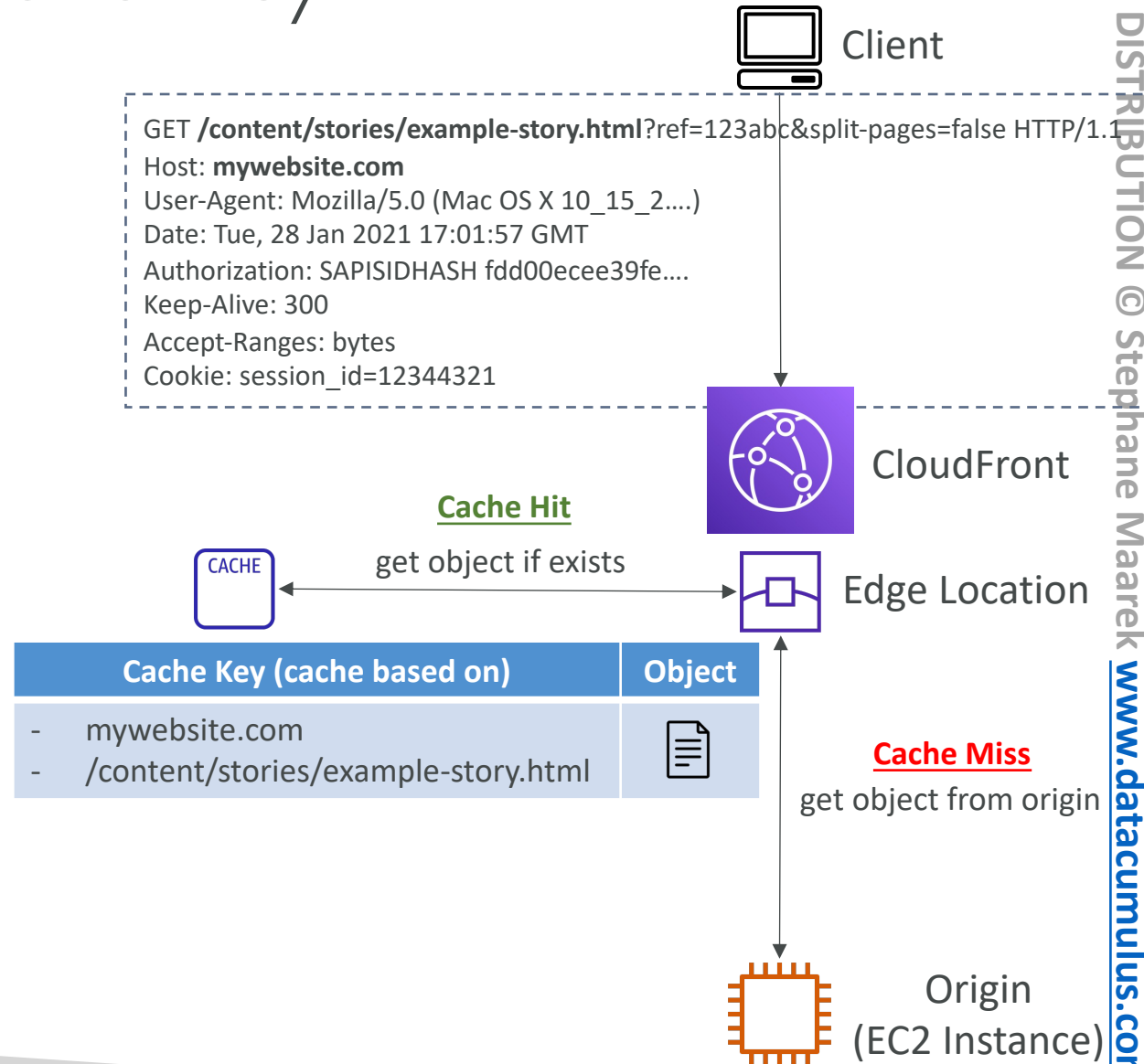
CloudFront Caching

- The cache lives at each CloudFront **Edge Location**
- CloudFront identifies each object in the cache using the **Cache Key** (see next slide)
- You want to maximize the Cache Hit ratio to minimize requests to the origin
- You can invalidate part of the cache using the **CreateInvalidation API**



What is CloudFront Cache Key?

- A unique identifier for every object in the cache
- By default, consists of **hostname + resource portion of the URL**
- If you have an application that serves up content that varies based on user, device, language, location...
- You can add other elements (HTTP headers, cookies, query strings) to the Cache Key using **CloudFront Cache Policies**



CloudFront Policies – Cache Policy

- Cache based on:
 - **HTTP Headers:** None – Whitelist
 - **Cookies:** None – Whitelist – Include All-Except – All
 - **Query Strings:** None – Whitelist – Include All-Except – All
- Control the TTL (0 seconds to 1 year), can be set by the origin using the **Cache-Control** header, **Expires** header...
- Create your own policy or use Predefined Managed Policies
- All HTTP headers, cookies, and query strings that you include in the Cache Key are automatically included in origin requests

CloudFront Caching – Cache Policy

HTTP Headers

GET /blogs/myblog.html HTTP/1.1
Host: mywebsite.com
User-Agent: Mozilla/5.0 (Mac OS X 10_15_2....)
Date: Tue, 28 Jan 2021 17:01:57 GMT
Authorization: SAPISIDHASH fdd00ecee39fe....
Keep-Alive: 300
Language: fr-fr



Client

request



CloudFront

- **None:**
 - Don't include any headers in the Cache Key (except default)
 - Headers are not forwarded (except default)
 - Best caching performance
- **Whitelist:**
 - **only specified headers** included in the Cache Key
 - Specified headers are also forwarded to Origin

CloudFront Cache – Cache Policy Query Strings

GET /image/cat.jpg?border=red&size=large HTTP/1.1

...



Client

request



CloudFront

- **None**
 - Don't include any query strings in the Cache Key
 - Query strings are not forwarded
- **Whitelist**
 - Only specified query strings included in the Cache Key
 - Only specified query strings are forwarded
- **Include All-Except**
 - Include all query strings in the Cache Key except the specified list
 - All query strings are forwarded except the specified list
- **All**
 - Include all query strings in the Cache Key
 - All query strings are forwarded
 - Worst caching performance

CloudFront Policies – Origin Request Policy

- Specify values that you want to include in origin requests **without** including them in the **Cache Key** (no duplicated cached content)
- You can include:
 - **HTTP headers:** None – Whitelist – All viewer headers options
 - **Cookies:** None – Whitelist – All
 - **Query Strings:** None – Whitelist – All
- Ability to add CloudFront HTTP headers and Custom Headers to an origin request that were not included in the viewer request
- Create your own policy or use Predefined Managed Policies

Cache Policy vs. Origin Request Policy

